



Giovanni Pettorru, Ph.D.

Contacts

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Languages

Italian: native
English: B2

EDUCATION

Secondary School Diploma | 2011–2016

ITC G.P. Chironi, Nuoro, Italia

- Economics sector, Administration, Finance and Marketing — specialisation in Business information systems.

Bachelor Degree — Computer Engineering | 2016–2020

University of Cagliari, Italy

- Final mark: 96/110
- Thesis: Implementation of a Magnetometer Based Vehicle Detection System for Smart Parking Applications.

Master Degree — Internet Engineering | 2020–2022

University of Cagliari, Italy

- Final mark: 110/110 cum laude
- Thesis: Secure and Low-Latency Communications Based on WebSocket over QUIC in Internet of Things Scenarios.

Doctor of Philosophy (Ph.D.) — Electronic and Computer Engineering | 2022–2026

University of Cagliari, Italy

- Final mark: cum laude
- Thesis: *Energy-Efficient and Trustworthy Location-Based Services for Next-Generation IoT*

Visiting Ph.D. Student | May 2024 – Sep 2024

Instituto Politécnico de Viana do Castelo, Portugal

PROFESSIONAL EXPERIENCE

Post-Doctoral Research Fellow | Apr 2026 – Present

University of Cagliari, Italy

Advanced research on drones as mobile localization and spatial reference platforms for spatial audio in immersive environments.

Tutor, teaching assistant | 2024–2026

University of Cagliari, Italy

Tutor for the *Network Security* course for Master's students in Internet Engineering.

Tutor, teaching assistant | 2025–2026

University of Cagliari, Italy

Tutor for *Cloud Networking* and *Digital Media* courses for Bachelor's students in Electronic, Computer, and Telecommunication Engineering.

Adjunct Professor | June 2025

University of Cagliari, Italy

Teaching activities for *Didactics for Telecommunication Systems* within the Initial University Training Program for Secondary School Teachers.

Freelance Worker | Nov 2025 – Apr 2026

University of Cagliari, Italy

Support for the development of an IoT-based platform for monitoring agricultural parameters.

Tutor, teaching assistant | 2021–2023

University of Cagliari, Italy

Tutor for the *Telecommunication Networks* course for Bachelor's students in Electrical, Electronic, and Computer Engineering.

Freelance Worker | Jul 2021 – Jan 2023

University of Cagliari, Italy

Support for the development of data acquisition algorithms on wireless networks and design of a monitoring system for alerting in critical urban situations.

OTHER ACTIVITIES & CERTIFICATES

Student Grant | Apr – Sep 2020

University of Cagliari, Italy

- M2M Comm — M2M Communication Standards in Smart Cities Environment

English B2 Level Certification | 2022

CLA, University of Cagliari, Italy

Training Course in European Project Design | 2023

Intellera Consulting S.p.a.

Specialized training in drafting and managing project proposals for 2021–2027 direct EU funding programs.

Certified Associate in Project Management (CAPM®) | 2026

Project Management Institute (PMI)

Certification of knowledge of PMI standards for project lifecycle management, planning, and control.

Research Mobility Grant (MGR) | 2026

Autonomous Region of Sardinia

Competitive international research mobility grant awarded following a merit-based evaluation of the research project proposal.

AWARDS

SERICS Lab2Bay | 2026

Fondazione SERICS @ INNOVIT

Selected for the acceleration program in San Francisco with the GreenShield project, addressing resource-aware IoT security.

Start Cup Sardegna | Oct 2025

University of Sassari, University of Cagliari and INNOIS

- Selected as one of the top 10 finalists in Start Cup Sardegna 2025.
- Special mention “Climate Change” awarded to the LENS project for its ability to integrate innovation and technology in the timely detection of extreme natural events.

Four-Minute-Thesis (4MT) | Dec 2025

IEEE ComSoc

Third place among the 20 finalists of the 4MT Competition by IEEE ComSoc at Globecom 2025.

AgriFuture | Jan 2025 – Jan 2026

Startup Desk by Sardegna Ricerche

Selected with AGROS among the top 10 high-tech innovative ideas for the agri-food sector.

Start Cup Sardegna | Oct 2024

University of Sassari, University of Cagliari and INNOIS

- Top 10 finalists
- Second prize (Spoke 7 of the e.INS) for the AGROS project.

Four-Minute-Thesis (4MT) | Dec 2024

IEEE ComSoc

Among the 15 finalists of the 4MT Competition by IEEE ComSoc at Globecom 2024.

Three-Minute-Thesis (3MT) | Jun 2023

IEEE ComSoc

Placed 3rd in the 1st Three-Minute Thesis (3MT) Competition by IEEE ComSoc at ICC 2023, where PhD students present their research to a non-specialist audience in three minutes.

PROFESSIONAL SERVICE & MEMBERSHIPS

Memberships

- IEEE Member (Institute of Electrical and Electronics Engineers) | 2021–Present
- IEEE Communications Society (ComSoc) | 2022–Present
- IEEE Young Professionals | 2021–Present
- IEEE Sensors Council | 2023–Present
- IEEE Internet of Things Community | 2023–Present

Program Committee (PC) Member | Nov 2023

AQ-IoT 2023: *Intelligent and Trustworthy Internet of Things Systems: Architectures and Qualities*.

Technical Programme Committee (TPC) Member

- ICCE Berlin 2026, *Special Session on Broadcasting*
- IEEE ICC 2027, *SAC — eHealth*

Web and Social Media Chair | 2026–Present

IEEE Communications Society (ComSoc) Italy Chapter.

Organizing Committee Member | Jul 2026

IEEE International Mediterranean Conference on Communications and Networking (MeditCom 2026).

Peer Reviewer for International Journals

- IEEE Communications Letters
- IEEE Transactions on Industrial Electronics
- IEEE Internet of Things Journal
- IEEE Transactions on Vehicular Technology
- IEEE Sensors Journal
- IEEE Transactions on Information Forensics and Security
- IEEE Access
- Inventions (MDPI)

PUBLICATIONS

Journal Articles

- [R1] **A Cross-Layer Survey on Secure and Low-Latency Communications in Next-Generation IoT**; M. Martalò, G. Pettorru, L. Atzori; *IEEE Transactions on Network and Service Management* (2024)
- [R2] **Trustworthy Localization in IoT Networks: A Survey of Localization Techniques, Threats, and Mitigation**; G. Pettorru, V. Pilloni, M. Martalò; *MDPI Sensors* (2024)
- [R3] **Balancing Complexity and Performance in Convolutional Neural Network Models for QUIC Traffic Classification**; G. Pettorru, M. Flumini, M. Martalò; *MDPI Sensors* (2025)
- [R4] **A Persistent and Secure Publish-Subscriber Architecture for Low-Latency IoT Communications**; G. Pettorru, M. Martalò; *IEEE Transactions on Network and Service Management (TNSM)* (2025)
- [R5] **A multimodal RSSI-based dataset for indoor navigation in interference-prone environments**; G. Pettorru, V. Pilloni, M. Martalò, S.I. Lopes; *Future Generation Computer Systems*, Elsevier (2026)
- [R6] **Robust Range-Based Localization Approaches Leveraging Multiple Wireless Interfaces**; G. Pettorru, V. Pilloni, M. Martalò; *IEEE Transactions on Wireless Communications (TWC)* (2026)

Conference Proceedings

- [C1] **Implementation of a Magnetometer based Vehicle Detection System for Smart Parking applications**; A. Floris, R. Girau, S. Porcu, G. Pettorru, L. Atzori; Int. Smart Cities Conference (ISC2), IEEE (2020)
- [C2] **Implementation of a Multisensors Fire-Fighting Monitoring System for Forest Protection**; G. Pettorru, M. Bertolusso, M. Spanu, M. Sole, M. Anedda, D. Giusto; CSCI, IEEE (2022)
- [C3] **A Machine Learning-based Approach for Vehicular Tracking in Low Power Wide Area Networks**; M. Bertolusso, M. Spanu, G. Pettorru, M. Anedda, M. Fadda, R. Girau, M. Farina, D. Giusto; BMSB, IEEE (2022)
- [C4] **A passive Wi-Fi based monitoring system for urban flows detection**; M. Bertolusso, G. Pettorru, M. Spanu, M. Fadda, M. Sole, M. Anedda, D. Giusto; IAICT, IEEE (2022)
- [C5] **Pedestrian and vehicular tracking based on Wi-Fi sniffing: a real-world case study**; M. Bertolusso, G. Pettorru, M. Spanu, M. Fadda, M. Sole, M. Farina, M. Anedda, D.D. Giusto; FITCE, IEEE (2022)
- [C6] **An IoT-based electronic sniffing for forest fire detection**; G. Pettorru, M. Fadda, R. Girau, M. Anedda, D. Giusto; ICCE, IEEE (2023) (*Speaker*)

- [C7] **A Hybrid WiFi/Bluetooth RSS Dataset with Application to Multilateration-Based Localization;** G. Pettorru, V. Pilloni, M. Martalò; MeditCom, IEEE (2023) (*Speaker*)
- [C8] **Using Artificial Intelligence and IoT Solution for Forest Fire Prevention;** G. Pettorru, M. Fadda, R. Girau, M. Sole, M. Anedda, D. Giusto; ICNC, IEEE (2023)
- [C9] **QUIC and WebSocket for Secure and Low-Latency IoT Communications: an Experimental Analysis;** G. Pettorru, M. Martalò; ICC, IEEE (2023) (*Speaker*)
- [C10] **Obfuscating Sensor-Based Activity Recognition in eHealth Applications: Is Encryption Enough Secure?;** F. Marcello, G. Pettorru, M. Martalò, V. Pilloni; ICC, IEEE (2024)
- [C11] **Preserving Privacy in CSI-based Human Activity Recognition: a Data Obfuscation Case Study;** F. Marcello, G. Pettorru, M. Martalò, V. Pilloni; GlobeCom, IEEE (2024)
- [C12] **Assessing the Interplay between IoT Localization Accuracy and the Two-Ray Channel;** C. Huidobro, M.G. Gaitán, C. Oberli, G. Pettorru, M. Martalò, V. Pilloni; WCNC, IEEE (2025)
- [C13] **How Do Jamming Attacks Impact the Performance of RSS-Based Localization Techniques?;** G. Pettorru, G. Nurcis, M. Martalò, V. Pilloni; ICC, IEEE (2025)
- [C14] **Multilateration-Assisted Fingerprinting-Based Localization for Dynamic IoT Environments;** G. Pettorru, V. Pilloni, M. Martalò; MeditCom, IEEE (2025) (*Speaker*)
- [C15] **A Reliability Index for Position Estimation in Trustworthy Location-Based Services;** G. Pettorru, A. Coni, V. Pilloni, M. Martalò; MeditCom, IEEE (2025) (*Speaker*)
- [C16] **Evaluation of Resource-Aware HTTP/3 Proxies for Smuggling Resilience in IoT Environments;** L. Pisu, G. Pettorru, R. Regano, D. Maiorca, G. Giacinto, M. Martalò; GlobeCom, IEEE (2025) (*Speaker*)
- [C17] **FiSH: Feature-informed Similarity Hashing for Edge-based Traffic Fingerprinting;** A. Mancosu, G. Pettorru, M. Martalò; ICC, IEEE (2026) (*Speaker*)
- [C18] **Near-Real-Time Traffic Classification in O-RAN: A Layer 2 Approach at the 5G Edge;** E. Pilia, G. Pettorru, M. Anedda, M. Murroni; ICCE, IEEE (2026)
- [C19] **A Feasibility Study of Edge-Centric Digital Twin for Autonomous Agricultural Decision Support;** F. Falchi, G. Pettorru, V. Pilloni, M. Martalò; CAFE, IEEE (2026)
- [C20] **On the Feasibility and Byzantine Resilience of WiFi HaLow Fingerprinting-Based IoT Positioning;** G. Pettorru, M. Curridori, M. Martalò, V. Pilloni; WF-IoT, IEEE (2026)
- [C21] **Lightweight and Tamper-Evident: A Decentralized Certification Pipeline for the Internet of Energy;** G. Nurcis, A. Coni, G. Pettorru, V. Pilloni, M. Martalò; WF-IoT, IEEE (2026)